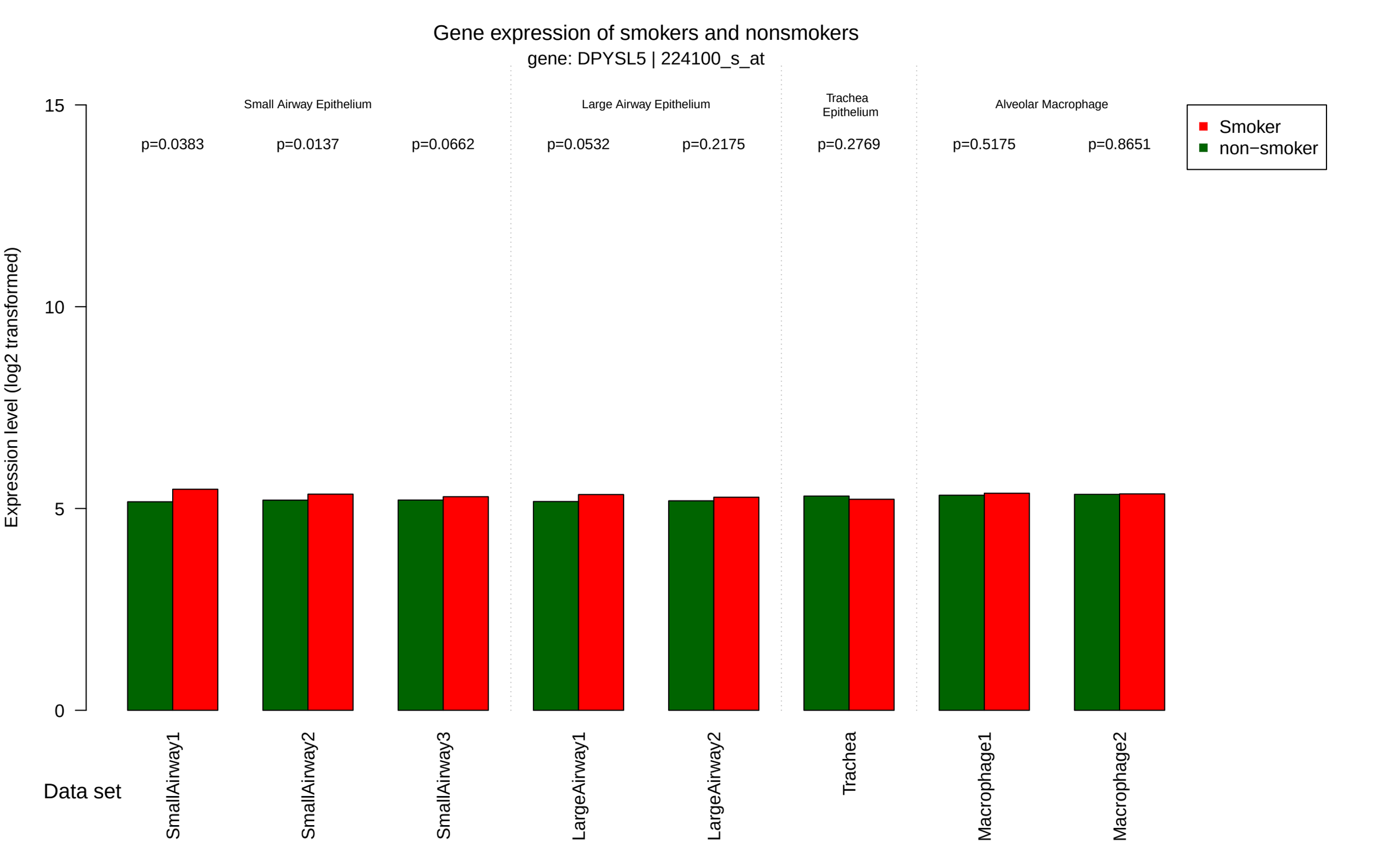
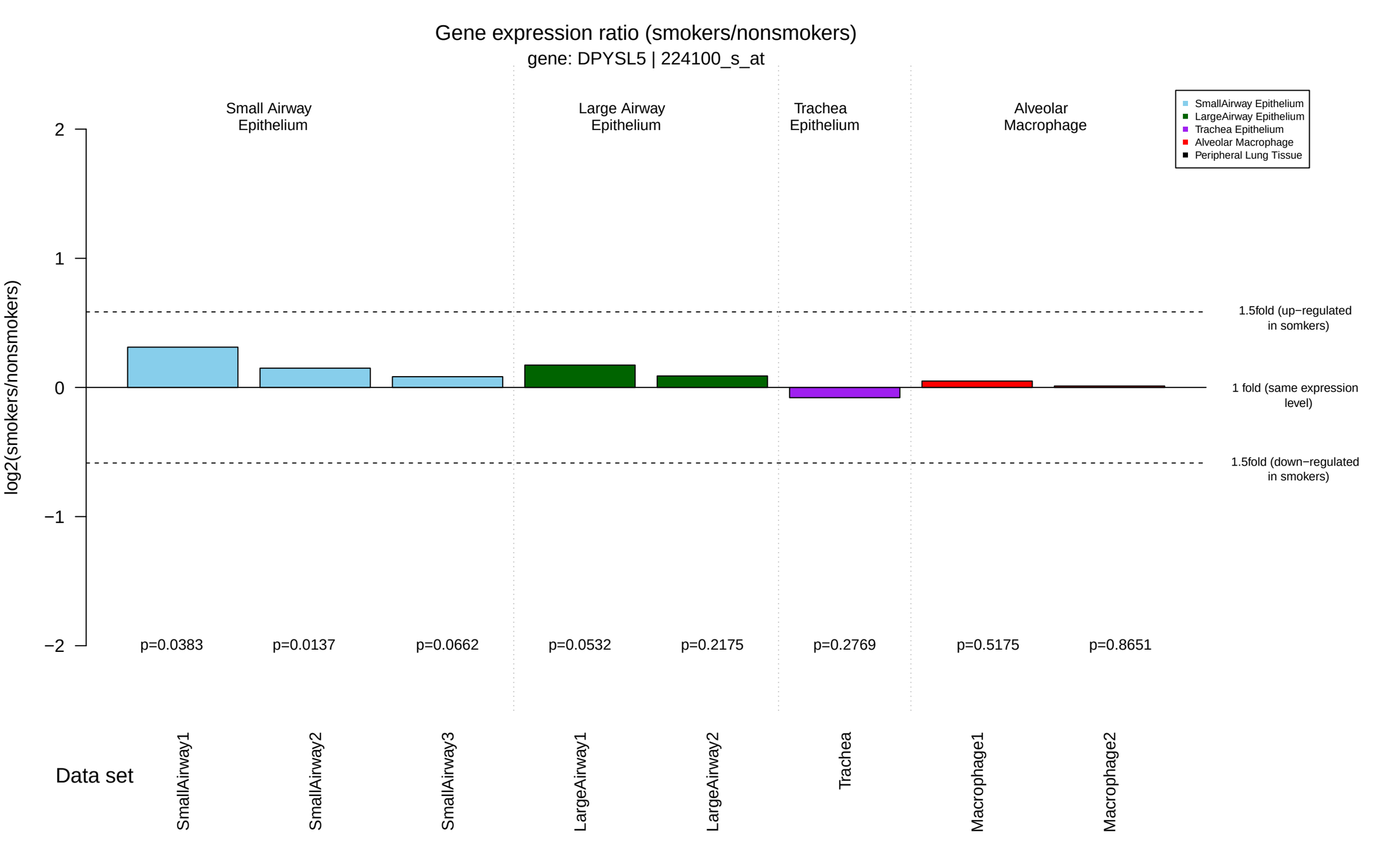
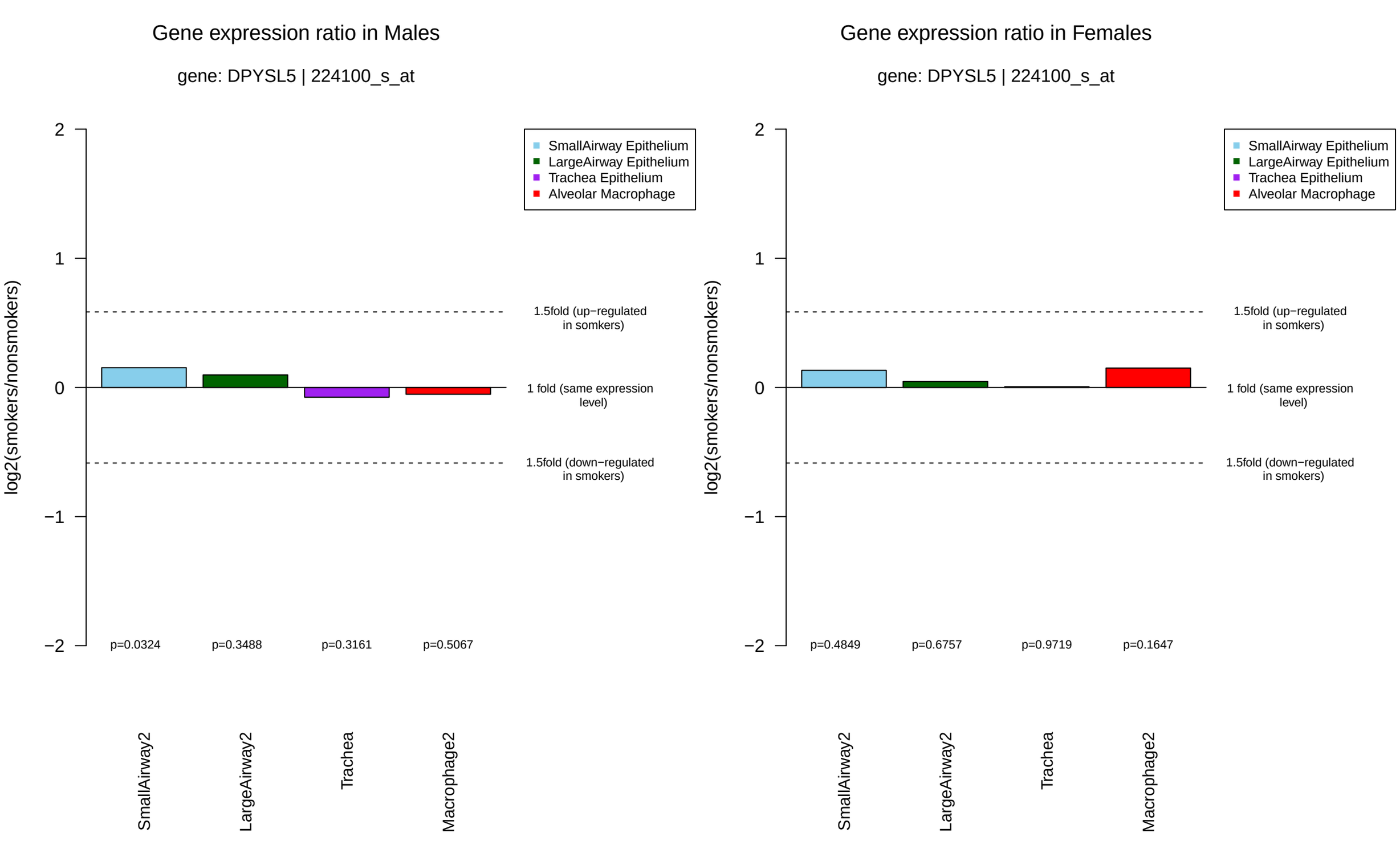
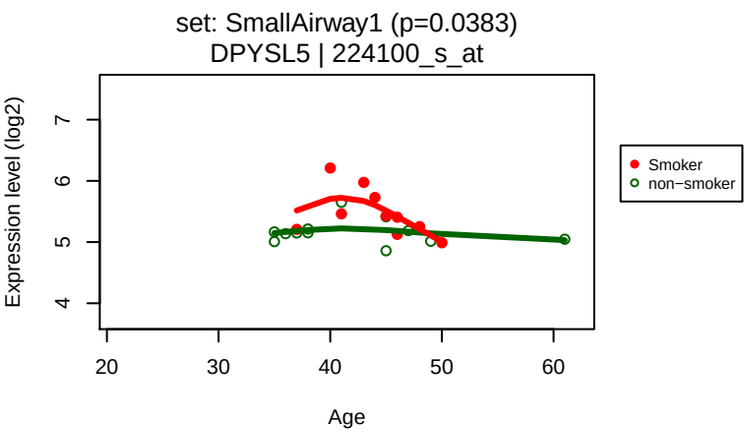
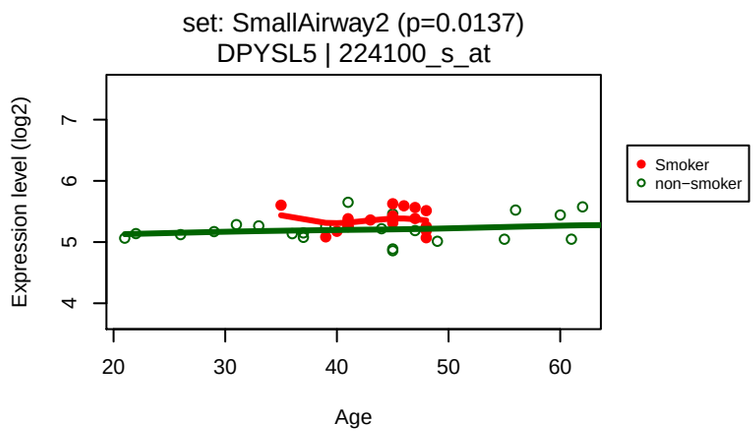
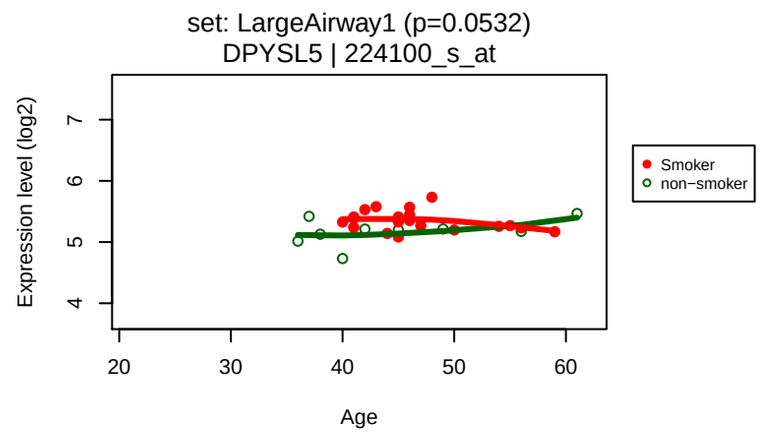

Gene expression of smokers and nonsmokers
gene: DPYSL5 | 224100_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0383
p=0.0137
p=0.0662
p=0.0532
p=0.2175
p=0.2769
p=0.5175
p=0.8651
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

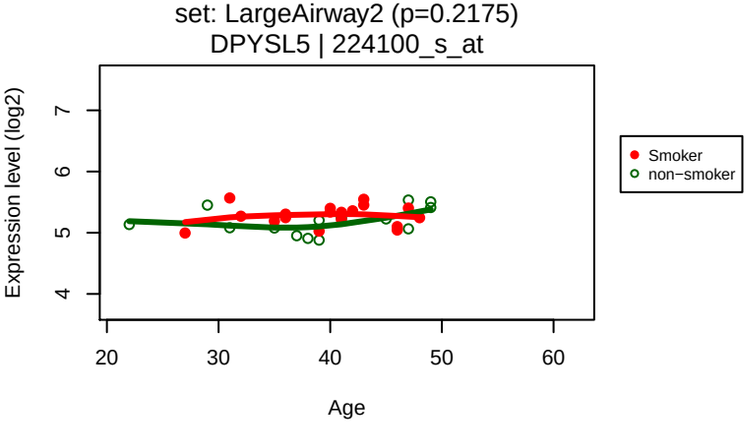

Gene expression ratio (smokers/nonsmokers)
gene: DPYSL5 | 224100_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0383
p=0.0137
p=0.0662
p=0.0532
p=0.2175
p=0.2769
p=0.5175
p=0.8651
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

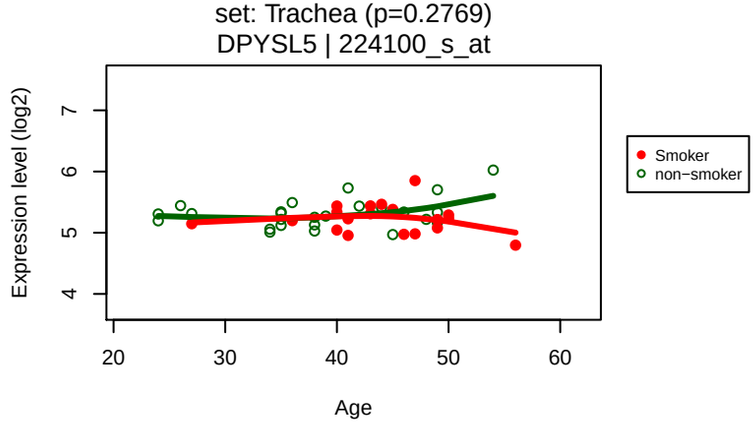

Gene expression ratio in Males
gene: DPYSL5 | 224100_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0324
p=0.3488
p=0.3161
p=0.5067
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DPYSL5 | 224100_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4849
p=0.6757
p=0.9719
p=0.1647
SmallAirway2
LargeAirway2
Trachea
Macrophage2

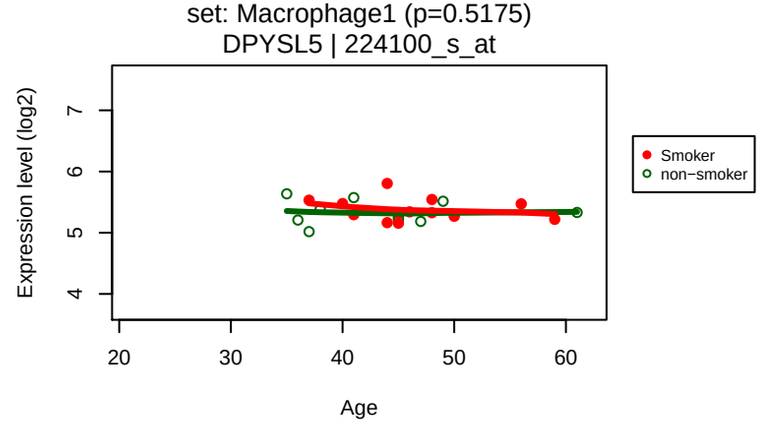

set: SmallAirway1 (p=0.0383)
DPYSL5 | 224100_s_at
Expression level (log2)
Age
Smoker
non−smoker

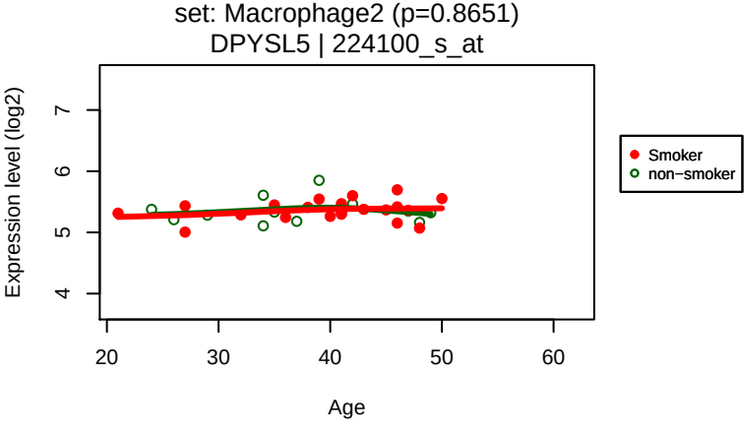

set: SmallAirway2 (p=0.0137)
DPYSL5 | 224100_s_at
Expression level (log2)
Age
Smoker
non−smoker

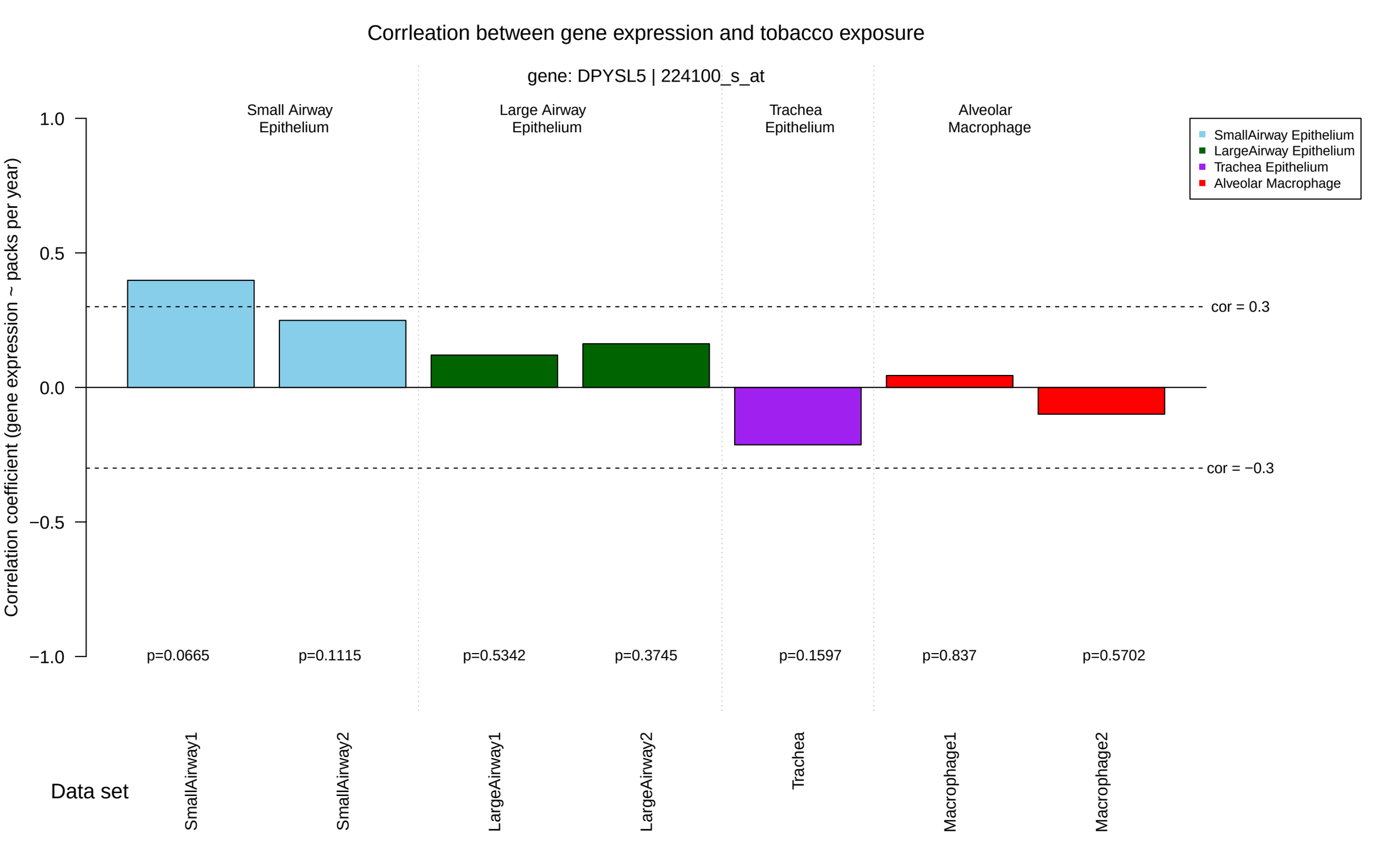

set: LargeAirway1 (p=0.0532)
DPYSL5 | 224100_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2175)
DPYSL5 | 224100_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2769)
DPYSL5 | 224100_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5175)
DPYSL5 | 224100_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8651)
DPYSL5 | 224100_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DPYSL5 | 224100_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0665
p=0.1115
p=0.5342
p=0.3745
p=0.1597
p=0.837
p=0.5702
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2